

NISHANK KOUL

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EDUCATION

PES University

Bachelor of Technology in Computer Science

Bengaluru, India

Dec 2021 – May 2025

TECHNICAL SKILLS

Languages: Python, Bash/Shell Scripting

Cloud Platforms: AWS, GCP, Azure

CI/CD & GitOps: Jenkins, GitHub Actions, GitLab CI, ArgoCD, Helm

Containers & Orchestration: Docker, Kubernetes (EKS, AKS, GKE)

Monitoring: Prometheus, Grafana, ELK Stack, CloudWatch, OpenTelemetry

Infrastructure as Code: Terraform, Ansible

Security: DevSecOps, SonarQube, Trivy, IAM, IRSA

EXPERIENCE

Vaani Research Labs | Founding DevOps Engineer (Bengaluru, India)

Sep 2025 – Present

- Architected a high-availability **AWS EKS cluster** handling **50,000+** daily calls with **Karpenter** and **KEDA** auto-scaling; reduced agent startup latency by **80%** utilizing a Kubernetes capacity buffer.
- Engineered traffic routing by configuring the **AWS Load Balancer Controller** with Ingress Grouping, sharing a single Application Load Balancer (**ALB**) across three domains with ACM-terminated TLS, reducing load balancer costs by **65%**.
- Established unified observability for **10+ microservices** using **OpenTelemetry**, **Prometheus**, **Grafana**, **Langfuse** for LLM tracing, and **Fluent Bit** for log routing, reducing system downtime by **15%** through faster root cause analysis.
- Implemented programmatic cost controls for **Generative AI services**, building Lambda functions to audit month-to-date **Amazon Bedrock** spend via Cost Explorer API and track third-party API quotas (OpenAI, Deepgram, ElevenLabs) with Slack alerting.
- Designed GitOps-driven CI/CD pipelines with **GitLab CI**, **Helm**, and **ArgoCD**, enabling **50+ deployments per week** with streamlined workflows.
- Led end-to-end **Azure to AWS migration**, including **AKS to EKS** and **PostgreSQL to RDS**, achieving zero data loss.

Stringify AI | Founding DevOps Engineer (Remote, USA)

Feb 2025 – Sep 2025

- Containerized and deployed cloud-native applications on **GCP** using multi-stage **Docker builds**, reducing image size by **30%**.
- Configured a **Global HTTP(S) Load Balancer** with Network Endpoint Groups (NEGs) and **CDN caching** to accelerate content delivery, reducing latency by **60%**.
- Provisioned a secure **PostgreSQL** database on GCP Compute Engine, restricting access via a **bastion host** and SSH tunneling.
- Streamlined CI/CD pipelines with **GitHub Actions**, accelerating deployment cycles by **40%** and ensuring consistency.
- Fortified security compliance by integrating **SonarQube (SAST)** and **Trivy** image scanning into CI/CD to reduce critical vulnerabilities by **40%**, and enabled **Cloud Audit Logging** for production admin audit trails.
- Achieved **35% cost optimization** on GCP VM spend via **Instance Schedules** and **Committed Use Discounts**.

Bimaplan | DevOps Engineer (Bengaluru, India)

May 2024 – Feb 2025

- Developed **Python** scripts for **AWS Lambda** to programmatically schedule EC2 shutdowns in Dev/UAT environments, reducing non-production cloud spend by **25%**.
- Codified **90%** of AWS infrastructure using **Terraform** with cross-region replication for **Disaster Recovery**, and deployed an **RDS read replica** to improve query performance by **40%**.
- Refined **Jenkins CI/CD pipelines** using Terraform to ensure **100% consistency** across environments; established backup strategies that cut rollback times by **60%**.
- Enforced **API Gateway rate-limiting** to prevent abuse, integrating **CloudWatch** alarms with **Slack** for real-time HTTP 429 alerting.

CERTIFICATIONS

GCP Certified Associate Cloud Engineer

Achieved May 2025

AWS Certified Cloud Practitioner

Achieved Oct 2024

RESEARCH & PUBLICATIONS

- Presented research on “**Overcoming the Challenges of Large Language Models: A Novel Proposition for Synthetic Data Validation**” at BDAI International Conference (IEEE), China, Jul 2024.
- Paper titled “**CounselAI: Transformative Career Counseling with GenAI**” accepted at SKIMA 16th International Conference (IEEE), United Kingdom, Jun 2025.